Bare Soil Mapping for Precision Agriculture

MENTOR

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PROBLEM & MOTIVATION

Why detecting bare soil matters at scale

Knowing exactly when and where agricultural soil is bare — free of crops, fully exposed — is one of the most valuable signals in modern agriculture. Yet there is no reliable way to detect this at scale across India.

- **Soil Organic Carbon (SOC) estimation** — measurable only when surface is visible
- **Crop calendar mapping** — harvest & sowing windows across regions
- **Drought & soil moisture monitoring** — bare fields as early stress indicators
- **Land degradation tracking** — fallow land, desertification, overuse
- **Carbon credit verification** — satellite-based, no costly field visits

THE GAP & OUR APPROACH

A problem that no existing tools are able to solve

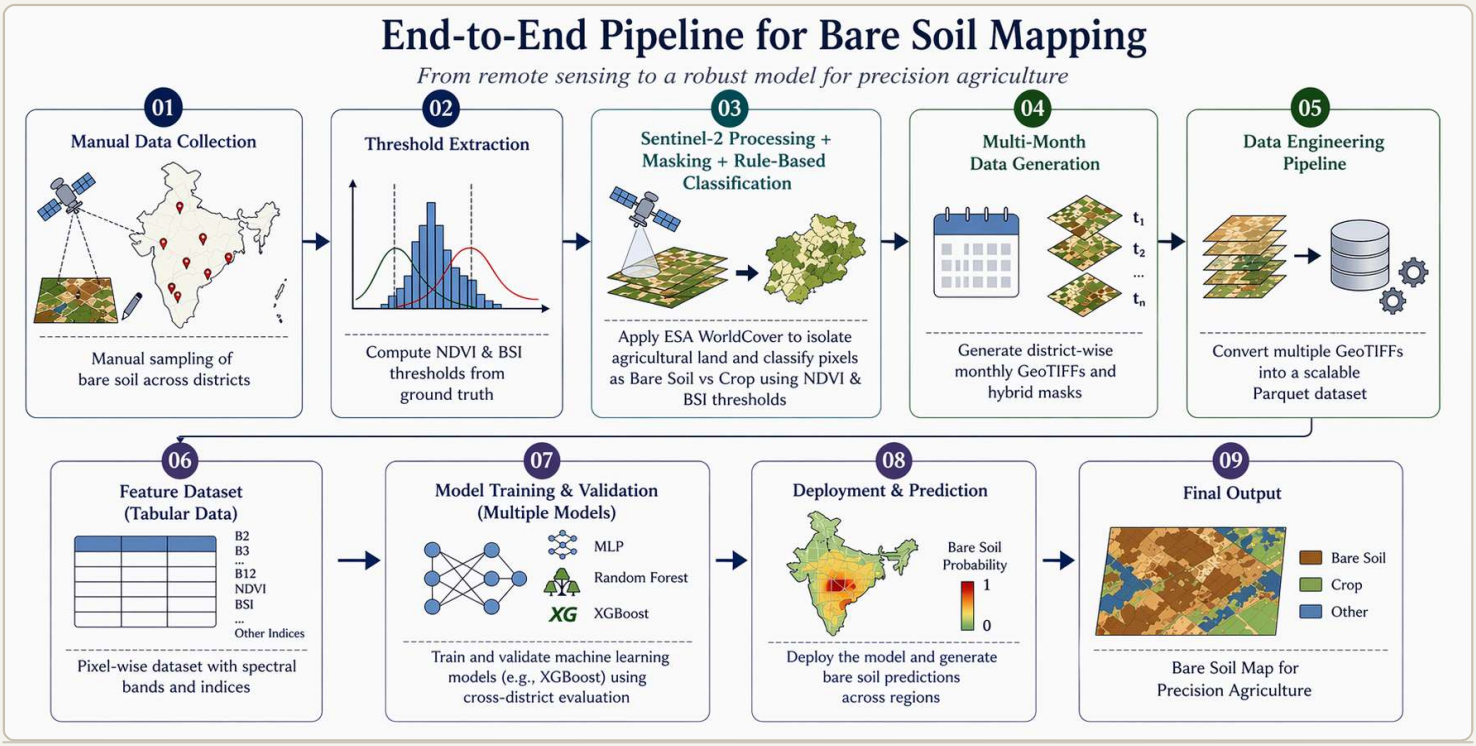
The project began with a direct goal: build an SOC prediction model. A pilot using drone imagery from Fatehgarh, Punjab quickly revealed a deeper problem.

No existing dataset or global model can reliably identify temporarily bare agricultural fields in India. Tools like Dynamic World and ESA WorldCover group bare farmland with deserts and construction sites — making precise detection impossible.

This gap became the core focus. Our work builds a scalable, data-driven bare soil detection pipeline trained across **49 districts** and **9 states** of India, using satellite imagery collected throughout **2024**.

HOW WE BUILT IT

End-to-end pipeline: from remote sensing to a deployable model



RESULTS — SAMPLE OUTPUT · AMRITSAR DISTRICT, NOVEMBER 2024

Sentinel-2 imagery vs. XGBoost bare soil prediction

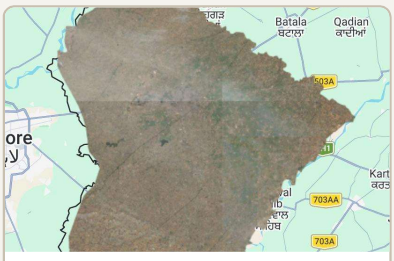


Fig 1. Sentinel-2 True Colour Composite
Raw satellite imagery — input to the inference pipeline.

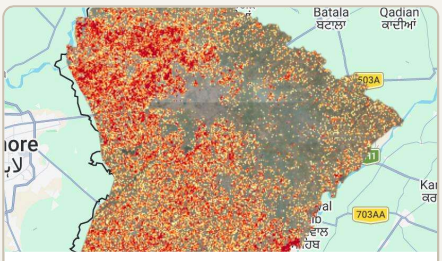


Fig 2. Bare Soil Prediction Map
Model output on cropland pixels — custom crop mask applied.

Colour Legend

Each pixel coloured by model prediction probability. Warmer = higher confidence bare soil.

Low	unlikely to be bare soil	50–60%
Moderate	uncertain / mixed pixels	60–70%
Good	probable bare soil signal	70–80%
High	strong likelihood	80–90%
Very high	highest probability of bare soil presence	90–100%

FINAL DEPLOYMENT

End-to-end web interface — region + date in, GeoTIFF out

User Input	GEE Export	Feature Computation	XGBoost Inference	GeoTIFF Output
Region + date via web interface	Sentinel-2 GeoTIFF fetched automatically	10 bands + 5 spectral indices	Per-pixel prediction + probability	Bare soil map ready for download

Steps highlighted in dark execute in the background — user interacts only with the input form.